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Intelligent Inputs Revolutionizing Agriculture: An Analytical Study

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3.1 INTRODUCTION

Agriculture plays a central role in the economy of every country. The need for food is increasing every day, along with the world population [1]. Currently, the traditional techniques of farmers cannot meet the demand. Various innovative methods of automation have been created to meet these needs, offering great career opportunities to many people in this field. Throughout human history, technology has long been used in agriculture to improve efficiency and reduce the labor intensity of agriculture. Since the invention of farming, humans and agriculture have evolved from better plows to irrigation, tractors, and modern artificial intelligence (AI) [2]. The increasing and more accessible availability of computer vision could be a major advance in this field. As significant global changes occur in our climate, ecology, and food demand, AI could transform 21st-century agriculture in the following ways:

- Improving time, labor, and resource efficiency
- Increasing the sustainability of the environment
- Smatter allocation of resources
- Providing real-time monitoring to encourage improved produce quality and health.

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For AI to achieve this, the agricultural sector must change. This calls for more technical and pedagogical investment in the agricultural sector to transform farmers' knowledge of their "field" into AI training. However, agricultural ingenuity and adaptation are zero innovative. For farmers, computer vision and agricultural robotics are the latest ways to use new technology to meet global food needs and improve food security. There is no doubt that today's work practices, agricultural yields, and quality exceed those of 500 or even 50 years ago. However, there is still a lot of room for improvement. By 2050, the planet will be home to 9.9 billion people, after which food demand is expected to increase from 35% to 56%. Not to mention how climate change is depleting natural resources, such as water and agricultural land. Fortunately, technology offers us another resource: AI. The agricultural sector is evolving at a whole new pace as advances in disease detection, predictive analytics, and computer vision technologies are used to monitor crops and soil. In addition to the potential, there is also rapidly growing interest and investment: Global "smart" agriculture, which includes machine learning and AI, is projected to triple to $15.3 billion by 2025, according to *Forbes*. Researchers predict that the market for AI in agriculture will grow with the combination. Compound annual growth rate (CAGR) of 20% to reach $2.5 billion by 2026. And that's just the beginning! This chapter examines the most innovative applications of AI that are reshaping the agricultural industry, including the following.

3.1.1 Crop and Soil Monitoring

The micro- and macronutrient composition of the soil directly affects the quantity, quality, and general health of the crop [3]. For farmers to maximize production efficiency, it is important to control the stages of the plants' development after planting them in the ground. Understanding the relationship between crop growth and the environment is critical to making changes that improve crop health. Now historically, human observations and assessments have been used to assess soil and plant health. However, this approach is neither accurate nor up-to-date. Instead, drones (unmanned aerial vehicles (UAVs)) are now used to capture aerial imagery and train computer vision models that can be used to intelligently monitor crop and soil conditions. Farmers can act quickly by learning specific problem areas using AI models.

3.1.2 Observing Crop Maturity

Such a labor-intensive task, which involves manually monitoring the growth phase of a wheat head, can be assisted by AI in the accuracy of agriculture. The researchers were able to build a "two-stage coarse to fine wheat ear detection mechanism" by collecting photographs of wheat at different stages of "orientation" over three years and under different lighting conditions. Farmers no longer had to go to the field every day to check their crops because this computer vision model identified the growth stages of the wheat more accurately than human observation. Another study looked at the accuracy of tomato ripeness detection using computer vision. The researchers developed an algorithm that tests the color of five different tomato ingredients and uses the results to determine the degree of ripeness of the tomato. The method attained a successful discovery and classification rate of 99.31%. Excessive care and overestimation of the development and maturity of crops require a lot of work from farmers. However, AI shows its ability to handle a significant part of the work easily and perfectly.

3.1.3 Hitting the Ground with Computer Vision

Returning to the significance of soil, a separate study examined the accuracy of computer vision images of soil structure and soil organic matter (SOM). Generally, farmers have to collect soil samples from the soil and take them to the laboratory for laborious and time-consuming evaluation [4]. Instead, the researchers decided to see if they could train an algorithm to perform the same task using image data from an inexpensive handheld microscope. The computer vision model was able to provide estimates of SOM and sand content that were as accurate as expensive laboratory processing. As a result, computer vision can not only eliminate a significant portion of the labor-intensive manual labor required in crop and soil monitoring but often does so more efficiently than humans.

3.1.4 Insect and Plant Disease Detection

We've seen how computer images powered by AI can identify and assess crop maturity and soil quality, but what about less predictable farming situations? Using image recognition technology based on deep learning, it is now possible to automatically detect plant diseases and pests. It works by creating models that can monitor plant health using image classification, recognition, and segmentation techniques.

3.1.5 Keeping Out the Bad Apples (Diagnosing Disease Severity)

A study on apple black rot – something we all definitely don't want on our apples! – gives a good example of this in practice. The Deep Convolutional Neural Network was trained using photographs of apple rot, which botanists classified into four levels of severity. The computer vision alternative, as in our previous cases, requires labor-intensive human search and evaluation. Luckily for agriculturalists, the AI model used in this study is 90.4% accurate in detecting and diagnosing the severity of the disease! In an additional study, the researchers went even further by using the advanced YOLO v3 algorithm to detect many diseases and pests on tomato plants. Using a digital camera and smartphone, the researchers photographed the presence of 12 different diseases or pests in nearby tomato greenhouses. The model was trained with images of different resolutions and object sizes, after which it was able to identify diseases and pests with 92.39% accuracy in less than 20.39 milliseconds.

3.1.6 Finding Bugs with Code

The analysts, to begin with, set up a sticky trap to capture six diverse sorts of flying creepy crawlies and take real-time photographs. They used YOLO object recognition for object detection and coarse computations and support vector machines (SVMs) using global features for object classification and fine computations. When all was said and done, their computer vision show was able to number them with 92.5% exactness and recognize between bees, flies, mosquitoes, moths, wasps, and natural product flies at 90.18%. Computer vision-based AI has the potential to monitor the health of our food systems, according to this study. In addition, it reduces labor inefficiencies while maintaining the accuracy of observations.

3.1.7 Livestock Health Monitoring

So far, we've mainly talked about crops, but farming includes more than wheat, tomatoes, and apples. Our agricultural system also relies heavily on animals, which require a little more management than plants. Can computer vision keep up with pigs, cows, and chickens? Cattle-Eye is a great example of agricultural AI. They use aerial cameras and computer vision algorithms to monitor livestock health and behavior. This means that the farmer does not necessarily have to stand next to the cow to see the problem. By remotely monitoring and tracking flocks in real time, farmers can be alerted as soon as problems are detected. Livestock aren't the only victims. Additionally, computer vision enables the following:

- Count animals, look for sickness, spot odd conduct, and keep an eye on significant activities, such as giving birth.
- Gather information using drones and cameras.
- Combine this with other technologies to educate farmers on animal health and food, as well as water availability.

3.1.8 Intelligent Spraying

UAVs equipped with computer vision. AI enables automatic and consistent application of pesticides or fertilizers across the field. Real-time detection of the target spray area enables UAV sprayers to work with high precision in terms of spray area and spray volume. As a result, there is significantly less chance of contaminating crops, livestock, people, and water systems. Although this place has a lot of potential, there are still challenges. For example, using multiple UAVs to spray a large area is much more efficient, but it can be difficult to determine specific mission cycles and flight paths for each vessel. However, this does not always mean intelligent injection is no longer an option [5]. Virginia Tech researchers have developed a smart spraying system based on servo-driven sprayers that use computer vision to identify weeds. A camera attached to the sprayer records the geographic location of the weed and assesses the size, shape, and color of each elusive plant to deliver the exact amount of herbicide to the target. In other words, it acts as a kind of herbicide. Unlike the Terminator, the computer vision system can shoot with such precision that it prevents additional damage to prey or the environment.

3.1.9 Automatic Weeding

The use of AI extends beyond weed control sprinklers. Other, more direct computer vision robots remove unwanted plants. While computer vision can spot an insect or a chicken behaving strangely, weed detection doesn't save the farmer a lot of work. For AI to be even more useful, it needs to find and remove weeds. Being able to remove weeds by hand not only saves a farmer's work but also reduces the need for pesticides, making the entire farming process much more environmentally friendly and sustainable.

3.1.10 Robots in the Weeds

With target detection, weeds and crops can be easily distinguished from each other. But the real power is seen when computer vision and machine-learning techniques are applied to the construction of weeding robots. All of this is a great introduction to BoniRob, an agricultural robot that uses cameras and image recognition technologies to look for weeds and then chase them out of the ground. It acquires the ability to differentiate between weeds and crops through visual training based on leaf size, shape, and color. In this way, BoniRob can remove unwanted plants from the field without having to worry about destroying anything irreplaceable. To accomplish this, researchers are creating agricultural robots for the detection of weeds and soil moisture. It can move around the field in this way, removing plants and drenching the soil with just the right amount of water. According to the test results, the plant classification and eradication rate of this technology is 90% or more, while the moisture content of the deep soil is maintained at 80%–10%.

3.1.11 Aerial Survey and Imaging

It's no surprise that computer vision has great applications for mapping land and monitoring crops and livestock. However, this does not diminish its importance for smart agriculture. AI can analyze satellite and drone footage to help farmers monitor their livestock and crops. They would be notified as soon as something unusual appeared, so they wouldn't have to constantly check the fields in person. Aerial photography improves the accuracy and efficiency of pesticide application. As mentioned earlier, ensuring that pesticides reach their intended destination saves costs and benefits the environment.

3.1.12 Produce Grading and Sorting

AI-based computer vision can continue to support farmers after harvest. With the help of imaging algorithms, defects, diseases, and pests can be detected during plant growth, and "good" products can be distinguished from defective or unpleasant products. Computer vision can automate sorting and sorting by analyzing the size, shape, color and quantity of fruits and vegetables with much greater accuracy and speed than qualified professionals.

3.1.13 Picture Perfect Produce

To distinguish between carrots with surface defects or carrots that are not the right size and shape, the researchers created an automatic method that sorts them. A "good" carrot must, therefore, have the correct shape ("convex polygon"). and must not have fibrous roots or surface cracks. Smart injection is no longer an option [5]. Virginia Tech researchers have developed a smart spraying system based on servo-driven sprayers that use computer vision to identify weeds. A camera attached to the sprayer records the geographic location of the weed and assesses the size, shape, and color of each elusive plant to deliver the exact amount of herbicide to the target. In other words, it acts as a kind of herbicide. In both cases, a huge amount of labor-intensive physical labor was saved. Figure 3.1 shows this clearly:

On the left side:

- Image of a field with weeds
- Camera attached to a servo-driven sprayer
- Arrows indicating data capture (location, size, shape, color)
- Herbicide spraying action

On the right side:

- Agricultural robots (depicted as robots with various tools)
- Crop soil monitoring (depicted as soil with sensors)
- Disease diagnosis (depicted as a plant with a magnifying glass)

Virginia Tech's smart spraying system utilizes computer vision for weed detection, optimizing herbicide use and reducing labor. The innovation integrates with other agricultural technologies for comprehensive farm management.

FIGURE 3.1
Smart spraying system by Virginia Tech-Greenhouse controlled environment agriculture.

Source: https://www.bing.com/images/search/yield-gruetzmacher-ai-automation-greenhouse-controlled-environment-agriculture

3.2 ANALYZING FARM DATA USING AI

It's amazing how farmers can now collect and analyze data from their fields due to AI; weather, temperature, water consumption, soil conditions, etc., can also be recorded in real time. Every day, thousands of field data points are recorded in the field. AI technology is already helping farmers improve yields through precision agriculture, which improves crop selection, hybrid seed selection, resource use, crop quality, and crop precision [6]. Precision agriculture uses AI technology to detect diseases, pests, and malnutrition in fields. AI sensors can detect weeds, target them, and choose which pesticides and herbicides to apply within a defined buffer zone [7]. This helps farmers maximize the number of pesticides and herbicides they use. In addition, AI helps farmers create seasonal forecasting models to increase crop production and accuracy. Using technology to predict weather trends in the coming months can also help farmers make smarter decisions. Small farms in less developed countries are the main target for seasonal forecasts, as information and data can be scarce. These small farmers must produce the best crops because they produce 70% of the world's crops. Farmers use AI-powered drone-based cameras in addition to topographic data to better photograph their produce and analyze site photos in real time to identify potential problems and areas for improvement. Drones monitor the product more efficiently and can cover a larger area than humans. The use of AI-based image recognition techniques for disease diagnosis, insect infestation detection, and plant identification is also becoming more widespread. A program based on X-Fito understanding was created by Cruz et al. (2017) to identify signs of Olive Quick Decline Syndrome (OQDS) [8]. The system uses transform-focused neural networks (DP-CNN) and new abstraction-level fusion techniques to improve diagnostic accuracy. Crus's PCA (Principal Component Analysis) system for industrial applications is used in a similar manner by farmers to make quick management decisions and reduce disease (Ibid).

3.2.1 Yield Management Using AI

AI, cloud technology, satellite imagery, and advanced analytics have created a modern smart farming ecosystem. By combining these technologies, farmers can increase average yields and better control prices. Cortana Intelligence Suite and machine learning are two advisory services Microsoft is now offering to farmers in Andhra Pradesh. This pilot project used AI applications in agriculture to provide data, soil preparation, soil test-based fertilization, seed treatment, ideal sowing depth, and more. Mobile robots and field sensors support digital farming robots, multifunction cameras, and laser scanners in blind spots and wireless areas.

3.2.1.1 Tackling the Labor Challenge

Due to the massive migration of labor to the city, farm operations are currently stalled. AI-powered robots improve employee productivity in a number of different ways. These robots can collect speedier and discover and expel weeds more precisely, lessening working costs and labor. Ranchers are, as of now, inquiring about chatbots to offer assistance [9]. Chatbots bolster agriculturists by replying to their questions and giving counsel and direction on particular agrarian issues.

3.2.1.1.1 Applications of AI in Agriculture

In the above image, AI technologies are depicted on the left side, showcasing their role in various aspects of agriculture, including crop yield production and price forecast, intelligent spraying, and providing productive insights. On the right side, AI is shown to facilitate connections with agriculture robots, crop soil monitoring, and disease diagnosis. This illustrates the comprehensive impact of AI in optimizing agricultural processes, from enhancing productivity to enabling precision management practices and proactive disease control.

As with traditional farming methods, farmers face many challenges. In this field, AI is often used to solve these problems (Figure 3.2). AI is a

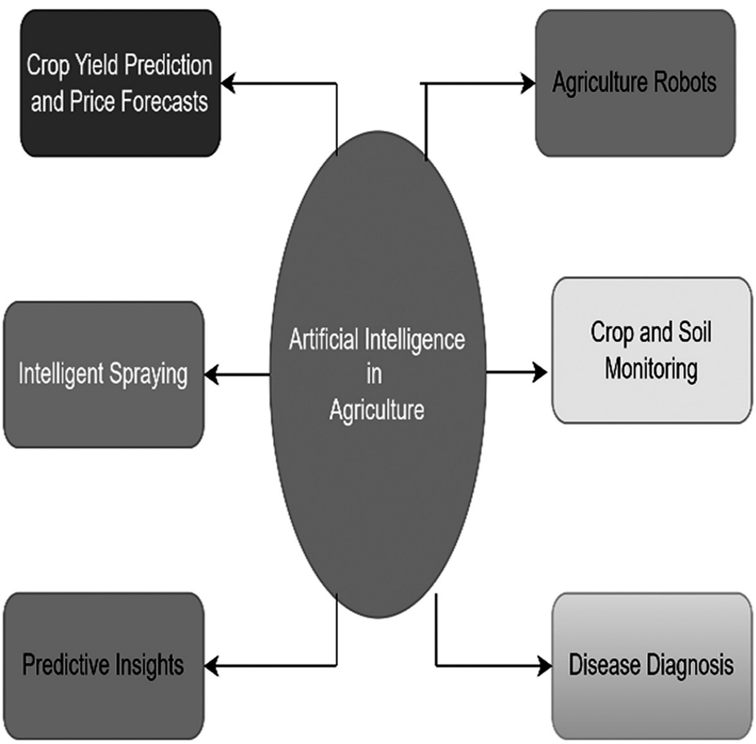

FIGURE 3.2
Applications of AI in agriculture.

Source: Prepared by the researchers

pioneering technology in agriculture today. This has benefits in producing healthier crops, pest control, soil monitoring, and more [10].

3.3 AI START-UPS IN AGRICULTURE

Based on data inputs, attempts are being made worldwide to automate the operations of agricultural businesses and use AI and related technologies to streamline agriculture [8]. This has led to the creation of countless start-ups.

- ***Prospera*: **The start-up has developed a cloud-based application that integrates all the data available to farmers, including water and soil sensor data, aerial photography, and more. It then connects to field devices that analyze the data and recommend desired results. Commonly used in greenhouses and outdoors, Prospera works with a variety of sensors and technologies, including computer vision. Input from these sensors is used to establish relationships between various data tags and their predictions.
- ***Blue River Technology*:** Robotics, AI, and computer vision are being combined to reduce costs and use of pesticides. Computer vision identifies each plant individually, and machine learning determines how to analyze each plant's characteristics. This way, the robot can intelligently use agricultural machinery and make the necessary decisions.
- ***Formbot*:** Since its founding in 2011, this organization has promoted precision farming techniques for smallholders and engaged the public in precision research. It allows farmers to perform a variety of tasks, including weed detection, soil testing, and plant watering, using robots running on an open-source software platform [11].
- ***Harvest CROO Robotics*:** This robotic strawberry picking system uses AI and machine vision to find and identify ripe berries so the strawberries can be harvested. Strawberry growers face a serious labor shortage, which raises growing costs and increases the likelihood of a subharvest. As a result of the additional use of AI and the development of robotic harvesting systems, the need for manual labor for producers would be reduced, logging costs would be reduced, and overall competitiveness would increase.

- ***Gramophone (Agstack Technologies)*:** Using image recognition skills, the right information, methods, and ingredients are available at the right time so farmers get the best possible harvest. The company predicts pests and diseases, predicts when food prices will increase production, and recommends products to farmers using AI and machine learning.
- ***Jivabhumi*:** Create an "intelligent" agricultural market to balance the often insufficient supply and demand of agricultural products [12]. A revolutionary food aggregation system that combines agricultural products, electronic marketing services, and innovation. Technologies such as blockchain are used to collect information about products at various points in the supply chain.

3.4 CHALLENGES IN THE ADOPTION OF AI

Despite the enormous promise of AI in agriculture, there are still many farms that do not have access to cutting-edge machine-learning techniques. The vulnerability of agriculture to external forces such as weather, soil quality, and pests is also crucial. Due to changes in external circumstances, a decision that seemed wise in the planning phase did not necessarily turn out to be the best. AI systems require large amounts of data to train algorithms to make accurate predictions. Finding temporal data for large agricultural areas is difficult, while collecting geographic data is easy. As the data infrastructure evolves, building a robust machine-learning model takes time. This is one reason why AI is not used in field solutions but in agricultural products, such as seeds, fertilizers, and pesticides. When it comes to farmers who are in control of their decisions and are able to make their own decisions and predictions depending on the circumstances, agriculture is still at an extremely early stage.

3.5 CONCLUSION

Summarizing the aforementioned research/study, we conclude that in all industries, including education, banking, robotics, agriculture, etc., the development of AI is considered one of the most important technologies.

This is changing the agricultural sector and has a major impact on it. AI protects the agricultural sector from a number of challenges, such as food security, population growth, climate change, and labor problems [13]. Modern agriculture and AI have both risen to unprecedented heights. AI has improved crop production, real-time monitoring, harvesting, processing, and marketing. A variety of high-tech computer systems are developed to detect several key criteria, including weed detection, crop detection, crop quality, and many others [14]. For applications to progress, they must be flexible to explore the many applications of AI in agriculture. It must be able to adapt to changing environments, support real-time decision-making, and obtain relevant information using appropriate platforms/structures. Another major drawback is that the cost of many options in the agricultural sector is very high. For the technology to be available at the farm level, the solutions must be more cost-effective. An open-source platform produces a more organized solution that allows farmers to deploy faster and better. The development of computer vision-based AI techniques requires a learning (preparing) approach that requires the collection and examination of a heap of illustrations in normal and energetic situations that precisely duplicate on-site conditions. AI technology usually advances by generating higher-quality data, allowing processors to deal with many imaging problems, such as inadequate lighting, inappropriate cropping, and misaligned objects [15]. Mobile devices can be combined with these algorithms and AI technology to provide an effective platform for pest and disease detection and pesticide mapping. The use and costs of pesticides can be significantly reduced, thus reducing their environmental impact, by using agrochemicals in the right amount and using pesticides only when necessary.

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